Dynamics and Learning for Interacting Particle Systems

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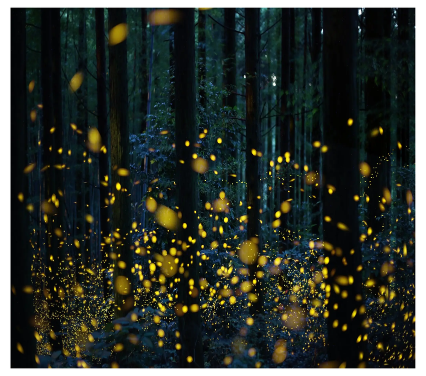
Introduction

Interacting Particle Systems in Nature

Collective behavior encompasses the study of how individuals within a group exhibit behaviors distinct from their actions in isolation. This field explores emergent patterns, spontaneous actions, and the influence of group dynamics on behavior. Understanding collective behavior sheds light on phenomena such as biological motion, rioting, and social behavior, offering insights into the complexities of group dynamics and societal interactions. The dynamics between agents can manifest via simple mechanisms, characterized by basic forces of attraction or repulsion, or in greater intricacy, arising from combinations of these fundamental interactions.



Theoretical Background

Estimate the interaction kernel ϕ of N agent system using M independent realizations.

$$\dot{\mathbf{x}}_i(t) = \frac{1}{N}\sum_{j=1}^{N}\phi(\| \mathbf{x}_j(t) - \mathbf{x}_i(t) \|)(\mathbf{x}_j(t) - \mathbf{x}_i(t)) \quad (1)$$

Obtain the estimator by minimizing the $\mathcal{E}$ over all the functions $\varphi \in \mathcal{H}_n$

$$\mathcal{E}_{L,M}(\varphi) = \frac{1}{LMN}\sum_{l,m,i=1}^{L,M,N} \| \dot{\mathbf{x}}_i^m(t_l) - \sum_{j=1}^{N}\frac{1}{N}\varphi(r_{i,j}^m(t_l))\mathbf{r}_{i,j}^m(t_l) \|^2 \quad (2)$$

$\mathcal{E}$:= Discrete error functional, L:= Number of time steps
M:= Number of independent realizations,
$\dot{\mathbf{x}}_i(t)$:= Velocity of i^{th} agent at time t

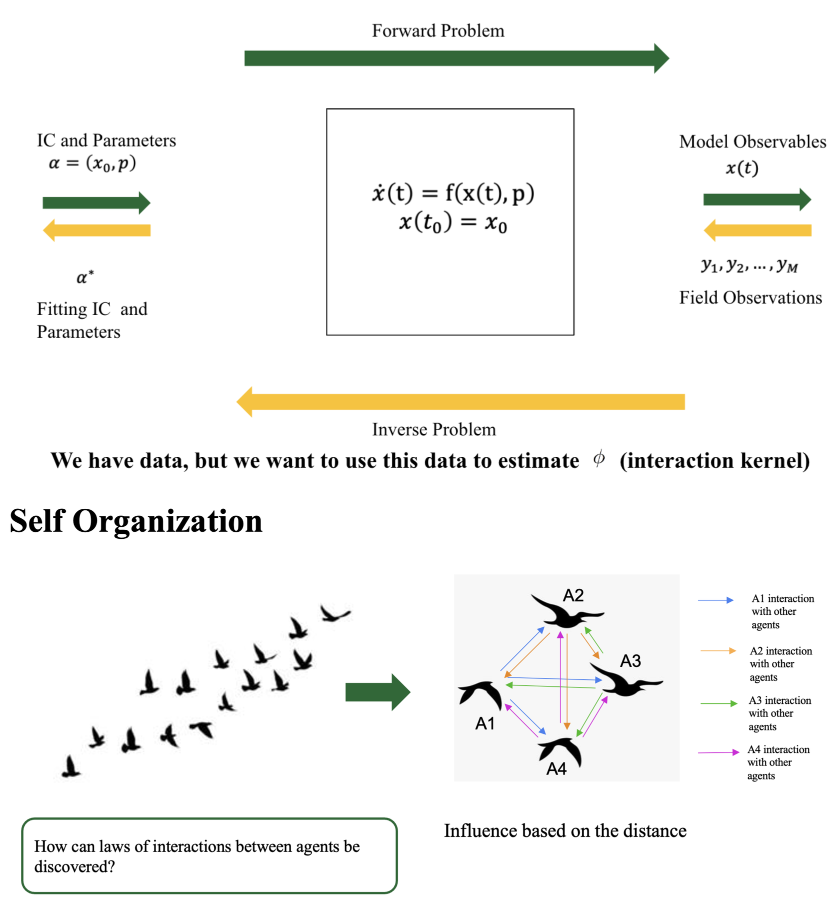
References

[1] Fei Lu, Ming Zhong, Sui Tang, and Mauro Maggioni. Nonparametric inference of interaction laws in systems of agents from trajectory data. *Proceedings of the National Academy of Sciences*, 116(29):14424–14433, July 2019.

[2] Francesco Bullo. *Lectures on network systems*, volume 1. Kindle Direct Publishing Seattle, DC, USA, 2020.

Implementation

Here we show the implementation of [1] which estimates the interaction kernel for first order deterministic systems.



Rewrite error functional using velocity observations $\mathbf{d}$ and kernel estimator $\dot{\varphi}$ as follows,

$$\mathcal{E}_{L,M}(\varphi) = \mathcal{E}_{L,M}(a) = \frac{1}{LMN}\sum_{m=1}^{M} \| \mathbf{d}^m - \Psi_L^m a \|^2 \quad (3)$$

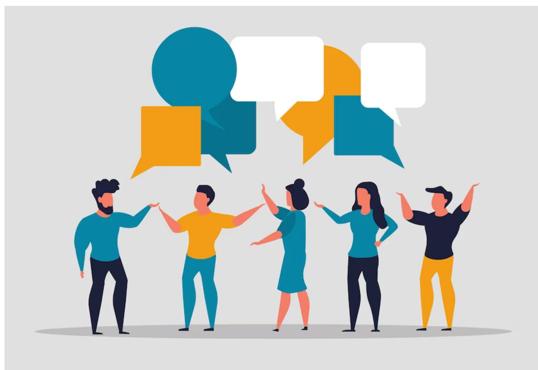
$\mathcal{H}_n$:= span $\{\psi_1, \psi_2, \ldots, \psi_n\}$
$\dot{\varphi} = a_1\psi_1 + a_2\psi_2 + \ldots + a_n\psi_n$

Goal is to find $a_1, a_2, \ldots, a_n$

Pairwise distance base models for collective behavior

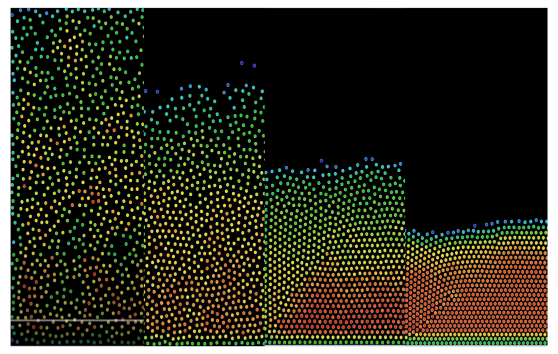
Opinion Dynamics

$$\phi(r) = \begin{cases} 1, 0 \le r < \frac{1}{\sqrt{2}} \\ 0.1, \frac{1}{\sqrt{2}} \le r < 1 \\ 0, 1 \le r \end{cases}$$



Lennard-Jones potential

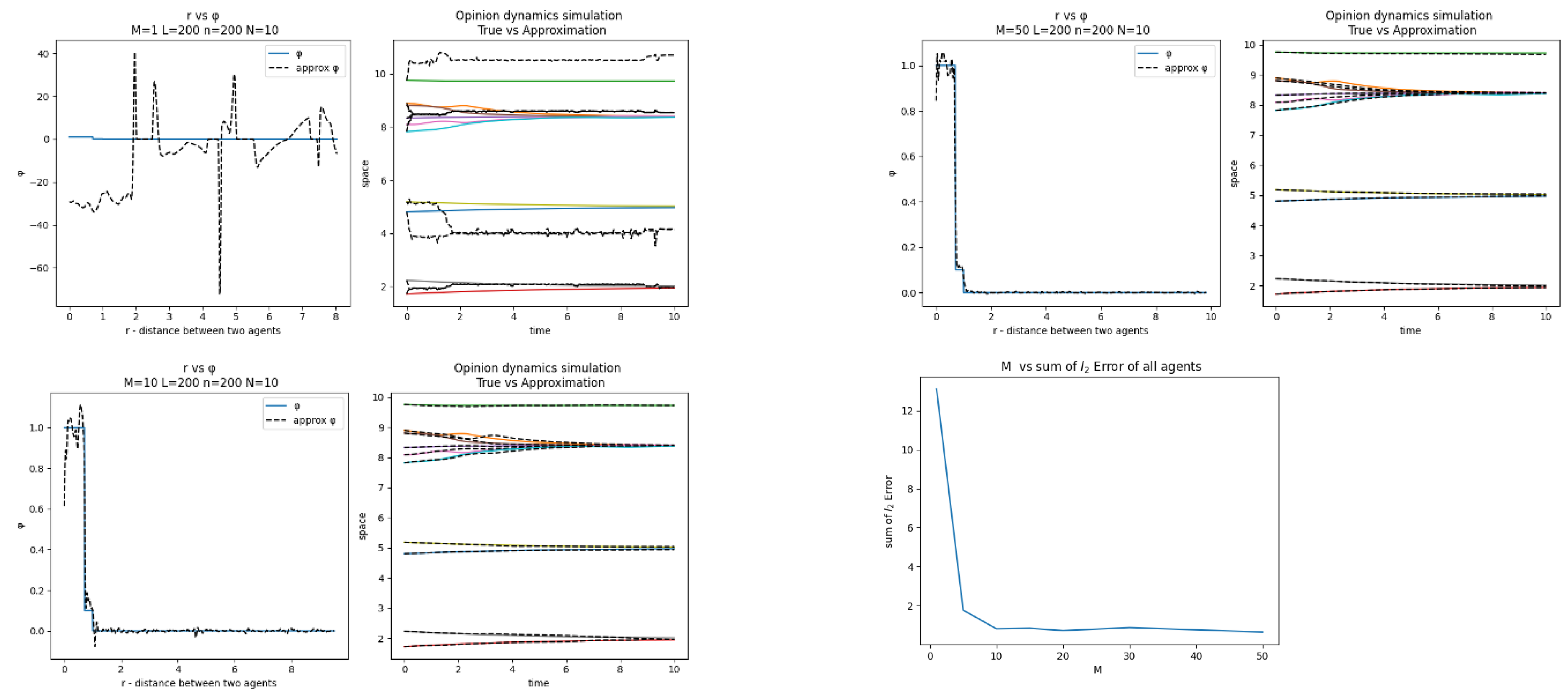
$$\Phi(r) = 4\epsilon[(\frac{\sigma}{r})^{12} - (\frac{\sigma}{r})^{6}]$$
$$= \epsilon[(\frac{r_m}{r})^{12} - (\frac{r_m}{r})^{6}]$$



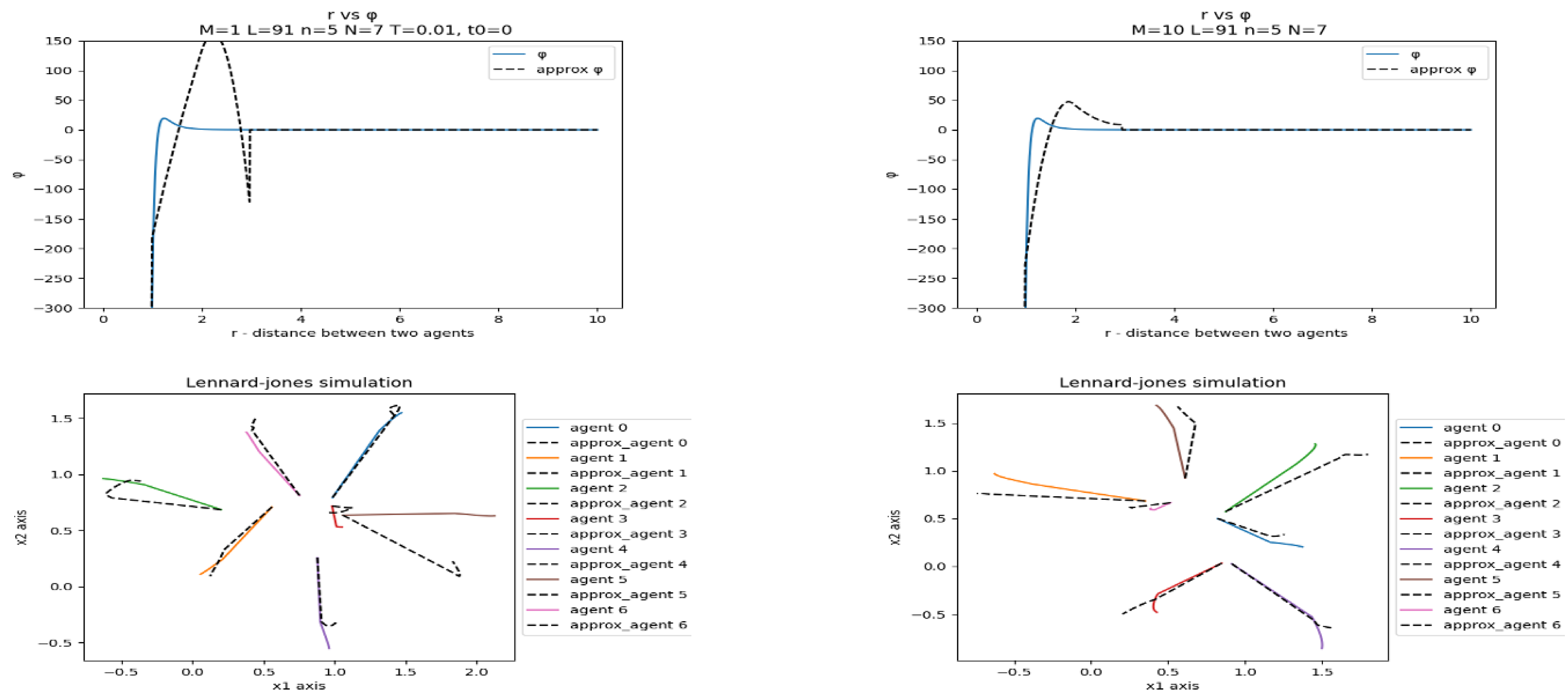
ϵ := The depth of the potential well.
σ := The finite distance at which the inter-particle potential is zero.
r_m := The distance at which the potential reaches its minimum.

Results

- Opinion Dynamics



- Lennard-Jones potential



Future Work

For our future work we plan to

- Recreating results of interaction kernel approximation for second order deterministic systems.
- Extend to systems which are second-order stochastic (and with state-dependent noise)
- Compare these methods which is extended from [1] to other standard inference methods for dynamical systems such as Sparse identification of nonlinear dynamics (SINDy) with non-linear library of functions customize for collective motion dynamics .
- Furtther with Neural Ordinary Differential Equations (NODE)